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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
not required.
Inclusion and Exclusion Criteria
not required.
Attrition
not required.
Sex as a biological variable
not detected.
Subject Demographics
Age: P5 stagger primer mix (0.5 mM) and uniquely barcoded P7 primer (0.5 mM) .
Randomization
not detected.
Blinding
not detected.
Power Analysis
not detected.
Replication
At 24 h post transfection , cells were incubated with replication restricted rVSVDG*G-GFP virus (Kerafast , EH1019-PM) at 5 MOI for 1 h at 37degC , 5 CO2 and the media was replaced with complete media .
Cell Line Authentication
not detected.
Data Information

Availability: Gene expression data used in this study are publicly available on the web portal and have been de-identified.	
Identifiers: Analysis of human lung scRNA-seq datasets Human lung scRNA-seq datasets from non-COVID-19 patients were accessed from the Human Lung Cell Atlas (https:github.comkrasnowlabHLCA) (Synapse syn21041850) , and from the Tissue Stability Cell Atlas (https:www.tissuestabilitycellatlas.org) (PRJEB31843)	PRJEB31843
Identifiers: Detailed clinical annotations of the GTEx cohort were obtained as controlled access data through dbGaP (phs000424.v8.p2) .	phs000424.v8.p2
Identifiers: To compare LRRC15 expression in the lungs of deceased COVID-19 patients with high vs low viral load 39 , we accessed the raw count data from GSE150316 and performed differential expression analysis using DESeq2 60 .	GSE150316
Identifiers: To compare LRRC15 expression in lung cell lines infected with SARS-CoV-2 vs mock controls 61 , we accessed the raw count data from GSE147507 and performed differential expression analysis using DESeq2 as above.	GSE147507
Identifiers: The coverslips were mounted on a slide using prolong(tm) glass antifade mountant with NucBlue(tm) (Invitrogen , P36981)	P36981

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
As a control , a same number of cells were stained with BV421 anti-hCD45 antibody (Biolegend , 368522) and the top 3 of the BV421-positive cells were sorted.	anti-hCD45		
Then the cells were washed two times and resuspended in FACS buffer containing the secondary antibodies at a 1:1000 dilution: AF647-labeled donkey anti-goat IgG (Invitrogen , A32849) or AF488-labeled goat anti-rabbit IgG (Invitrogen , A32731) .	AF647-labeled donkey anti-goat IgG		
	anti-rabbit IgG		
	A32731		
The plates were then incubated with goat anti-human IgG Fc secondary antibody , HRP (Thermo , A18817) at a 1:3000 dilution for 1 hour at room temperature.	anti-human IgG		
HRP-conjugated goat anti-human IgG Fc secondary antibody was used to detect the bound ACE2 .	ACE2		
To compare GFP expressions in ACE2- and LRRC15-positive cells , pseudovirus-infected cells were stained for surface ACE2 and LRRC15 as described above with following secondary antibodies: AF405-labeled donkey anti-goat IgG (Invitrogen , A48259) and PE-labeled donkey anti-rabbit IgG (Jackson ImmunoResearch , 711-116-152) .	LRRC15		
	AF405-labeled donkey anti-goat IgG		
	PE-labeled donkey anti-rabbit IgG		
Cells were subsequently incubated with recombinant anti-LRRC15 antibody (Abcam , ab150376) and SARS-CoV-2 spike antibody 1A9 (GeneTex , GTX632604) at 1:100 , followed by incubation with 1:500-diluted Alexa Fluor 555 conjugated goat anti-rabbit IgG antibody (Abcam , ab150078) and 1:200-diluted Alexa Fluor 488 conjugated goat anti-mouse IgG antibody (Abcam , ab150117) for 60 min at room temperature.	goat		
	IgG antibody (Abcam , ab150078) and 1:200-diluted Alexa Fluor 488 conjugated goat anti-mouse IgG antibody (Abcam , ab150117) for 60 min at room		
	anti-mouse IgG		
Experimental Models: Cell Lines			
MATERIALS AND METHODS Cell culture HEK293T (CRL-3216), HeLa	HEK293T		Suggestion: RRID:CVCL_0063(link)

(CCL-2), and A375 (CRL-1619) cells were purchased from ATCC.			
For A375 cells , 5 mgmL blasticidin (Gibco , A1113903) and 1 mgmL puromycin (Gibco , A1113803) were added as appropriate.	A375		Suggestion: RRID:CVCL_0132(link)
For HeLa cells , 5 mgmL blasticidin , 0.7 mgmL puromycin , 200 or 400 mgmL hygromycin (Gibco , 0-687-010) , and 100 mgmL zeocin (Thermo , R25001) were added as appropriate.	HeLa		Suggestion: RRID:CVCL_0030(link)
7.8 x 107 A375-dCas cells were transduced with the CRISPRa library at 0.3 MOI to make 2.4 x 107 transduced cells , which is sufficient for the integration of each sgRNA into 500 cells .	A375-dCas		
For 1:1 ratio co-culture , 1 x 104 HeLa-ACE2 cells and 1 x 104 HeLa or HeLa-sgLRRC15 cells were plated per well .	HeLa-ACE2		
Microscopic analysis 8 x 103 HeLa-ACE2 cells and 3.2 x 104 HeLa or HeLa-sgLRRC15 cells (1:4 ratio) were co-plated per well in 8-well chamber slides (Nunc) .	HeLa-sgLRRC15		
Recombinant DNA			
Generation of genetically modified cell lines Individual sgRNAs (sgLRRC15 1: GACATGCAGGCACTGCACTG sgLRRC15 2: AGTGTCAGCCCGGGACATGC sgACE2: GTTACATATCTGTCCTCTCC) targeting the candidate genes were cloned into linearized pXPR_502 (Addgene , 96923) for CRISPR activation .	pXPR_502		Suggestion: RRID:Addgene_96923(link)
A375-dCas or HeLa-dCas cells were generated by transducing with pLenti-dCas9-VP64-Blast (Addgene , 61425) .	pLenti-dCas9-VP64-Blast		
Stable ACE2 expressing HeLa cells (HeLa-ACE2) were generated by transducing HeLa-dCas cells with pLENTI_hACE2_HygR (Addgene , 155296) followed selection with hygromycin .	pLENTI_hACE2_HygR		Suggestion: RRID:Addgene_155296(link)
For ectopic expression of LRRC15 , a lentiviral vector pCDH-MSCV-T2A-Puro (System Biosciences , CD522A-1) was modified to	pCDH-MSCV-T2A-Puro		

enable zeocin selection instead of puromycin .			
A codon-optimized LRRC15 ORF was cloned into pCDH-MSCV-T2A-Zeo with a C-terminal 3xFLAG tag and used to transduce HeLa-ACE2 followed by zeocin selection .	pCDH-MSCV-T2A-Zeo		
Expression vectors for SARS-CoV-2 Wuhan-Hu-1 (Addgene , 149539) , SARS-CoV-2 B.1.167.2 (Addgene , 172320) , SARS-CoV-2 B.1.1.7 (Addgene , 170451) , SARS-CoV-2 B.1.351 (Addgene , 170449) , SARS-CoV-2 P.1 (Addgene , 170450) , SARS-CoV-1 (Addgene , 170447) , MERS-CoV (Addgene , 170448) and VSV-G (Addgene , 12259) were used.	VSV-G		
Software and Algorithms			
For A375 cells , 5 mgmL blasticidin (Gibco , A1113903) and 1 mgmL puromycin (Gibco , A1113803) were added as appropriate.	Gibco		
After 30 min incubation at 4degC , the cells were washed two times , fixed with 4 formaldehyde for 15 min and washed and resuspended in FACS buffer before analyzing by flow cytometry using FACSCelesta (BD Biosciences) or Cytek	Cytek		
Data was analyzed with Flowjo software.	Flowjo		Suggestion: FlowJo, RRID:SCR_008520(link)
GraphPad Prism 9 software was used to perform nonlinear regression curve-fitting analyses of binding data to estimate dissociation constants (KD) .	GraphPad		Suggestion: GraphPad Prism, RRID:SCR_002798(link)
Anti-VSV-G (Sigma , MABF2337) was added at final concentration of 1 mgmL to neutralize residual rVSVDG*G .	Sigma		Suggestion: sigma-tau (RRID:SCR_000488)(link)
To compare LRRC15 expression in lung cell lines infected with SARS-CoV-2 vs mock controls 61 , we accessed the raw count data from GSE147507 and performed differential expression analysis using DESeq2 as above.	DESeq2		Suggestion: DESeq, RRID:SCR_000154(link)
Statistical analysis Statistical significance was determined using GraphPad Prism 9 software.	GraphPad Prism		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
For the Melms dataset , we also compared the relative proportions of each of the fibroblast subpopulations among total fibroblasts , using a two-tailed unpaired Mann-Whitney test to assess statistical significance.	Mann-Whitney test
For analysis of genes correlated with LRRC15 expression in the GTEx dataset , we computed the Spearman correlation between LRRC15 and all other genes represented in the GTEx dataset.	Spearman correlation
We performed multiple-hypothesis correction by the Benjamini-Hochberg method.	Benjamini-Hochberg
For visualization purposes , we further extracted the normalized expression values for LRRC15 in each of the samples and tested for statistical significance by two-tailed unpaired t-test .	t-test
Experiments were analyzed by one-way or two-way ANOVA with Dunnetts multiple comparisons test or unpaired two-pair t test as indicated.	ANOVA
	Dunnetts multiple comparisons test
	t test
Oligonucleotides	
Generation of genetically modified cell lines Individual sgRNAs (sgLRRC15 1: GACATGCAGGCACTGCACTG sgLRRC15 2: AGTGTCAGCCCGGGACATGC sgACE2: GTTACATATCTGTCCTCTCC) targeting the candidate genes were cloned into linearized pXPR_502 (Addgene , 96923) for CRISPR activation .	GACATGCAGGCACTGCACTG
	AGTGTCAGCCCGGGACATGC
	GTTACATATCTGTCCTCTCC

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